

Project Overview

- **Goal:** Predict house prices using linear regression and ensemble models.
- **Architecture:** Features a modular machine learning pipeline split across three distinct Jupyter Notebooks covering data loading, data cleaning, and model building.
- **Language:** The codebase consists entirely of 100% Jupyter Notebook file configurations

Key Evaluation Findings

- **Feature Impact:** Home area, bathroom counts, air conditioning presence, and total stories dictate market valuations the most.
- **Model Baseline:** The standard Linear Regression model captures exactly 64.05% of the overall market variance while Random Forest explain 61% of the overall market Variance.
- **Data Anomaly:** Predictions are heavily skewed at the high end due to multi-million dollar luxury outlier properties.
- **Recommendation:** Prioritize installing air conditioning systems before listings since it maintains a strong 0.45 correlation with pricing increases.
- **Efficiency:** In this dataset some house have high price acts as outliers and decreasing the efficiency and performance of the model, so Linear Regression or Random forest is not good model for the given datasets.